

Sampling I: from uniform to any distribution

A seeded formula makes uniform numbers. The inverse CDF, Box-Muller and a Cholesky factor reshape them into the draws you need.

THE QUESTION

A computer makes only uniform numbers. How do we get exponential, categorical or correlated Gaussian draws from them, and how far can we trust averages over the draws?

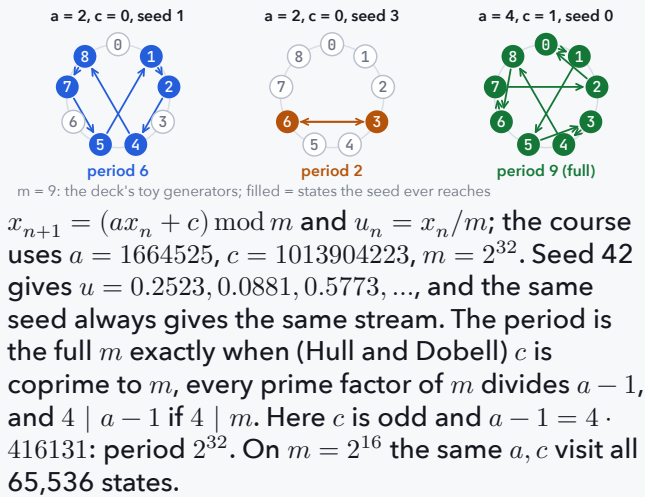
THE INTUITION

A seeded recurrence makes the uniforms. Pass u through F^{-1} for any 1D distribution. Box-Muller makes two normals from two uniforms, and $\mu + Lz$ correlates them.

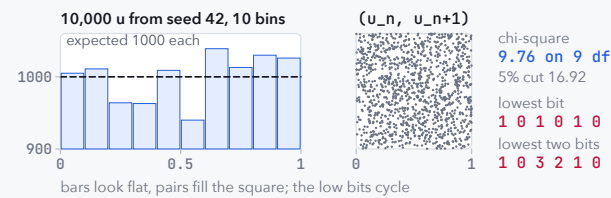
KEY TAKEAWAY

$x = F^{-1}(u)$ for any 1D law; Box-Muller and $\mu + Lz$ for Gaussians. A Monte Carlo mean has error $\sigma_f/\sqrt{N}$ in any dimension.

1 An LCG is a seeded recurrence, so it cycles

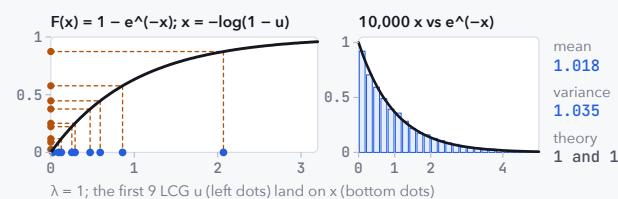


2 Flat bins and clean pairs, but weak low bits



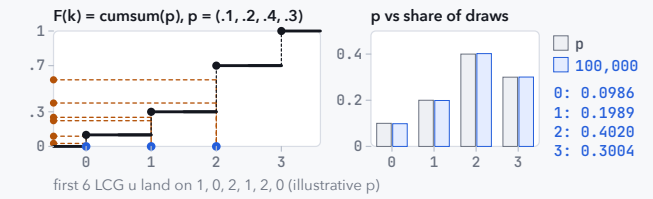
Ten bins of 10,000 u : $\chi^2 = \sum (O - E)^2/E = 9.76$ on 9 df, below the 5% cut ($p \approx 0.37$). The notebook's $U(-2, 2)$ counts, from $a + (b - a)u$, come out exactly (1006, 1012, ..., 1026): the generator is deterministic. Lag-1 correlation: $-6 \cdot 10^{-5}$. With a, c odd, $x_n \bmod 2$ alternates, so use the top bits (x_n/m). NumPy's `default_rng` uses the stronger PCG64.

3 Inverse CDF: solve $u = F(x)$ for x



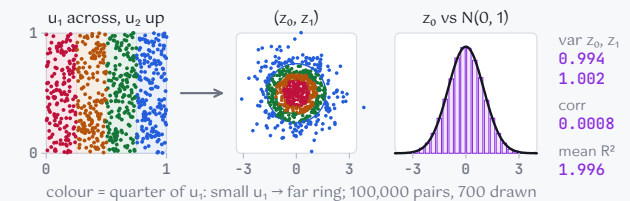
For continuous increasing F : $P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$. Exponential: $F(x) = 1 - e^{-\lambda x}$, so $x = -\log(1 - u)/\lambda$ (or $-\log u/\lambda$, as $1 - U \sim U(0, 1)$). The first five u give 0.2908, 0.0923, 0.8610, 0.2517, 0.4711; $u = 0.5$ gives the median $\log 2$.

4 Discrete: the first cumulative sum above u



For a categorical p , F is a staircase and $F^{-1}(u) = \min\{k : F(k) > u\}$: `np.searchsorted(np.cumsum(p), u, side="right")`, a binary search. Φ^{-1} has no closed form; the normal notebook uses $\sqrt{2}\text{erf}^{-1}(2u - 1)$ ($\Phi^{-1}(0.975) = 1.960$).

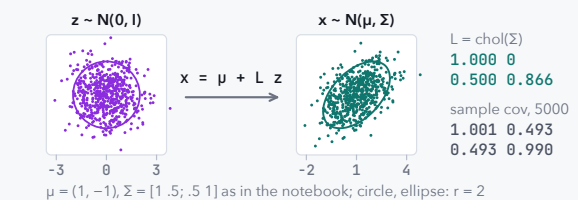
5 Box-Muller: two uniforms, two independent normals



$R = \sqrt{-2 \log U_1}$, $\Theta = 2\pi U_2$, $Z_0, Z_1 = R \cos \Theta, R \sin \Theta$

Two independent $\mathcal{N}(0, 1)$ have density $\frac{1}{2\pi} e^{-r^2/2} r dr d\theta$: Θ is uniform and $R^2 \sim \text{Exp}(1/2)$, whose inverse CDF is $-2 \log U_1$ (panel 3). 100,000 pairs: $P(|Z_0| > 1.96) = 0.049$. Put $1 - U_1$ in the log: a generator can return 0. $\mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$; the deck's $\mathcal{N}(\mu, \sigma)$ uses the sd.

6 A Cholesky factor adds the correlation



If $z \sim \mathcal{N}(0, I)$, then $x = \mu + Lz$ has covariance $LL^T = \Sigma$; here $L_{22} = \sqrt{1 - \rho^2}$. The notebook's 5000 draws (quoted) give variance 1.053. Its sd is $\sqrt{2/4999} = 0.020$, so that is 2.7 sd high: 0.85% of reruns do it. Any L with $LL^T = \Sigma$ works; Cholesky needs Σ positive definite (add $10^{-6}I$ if nearly singular) and costs $d^3/3$.

Average the draws, and report the error

Any expectation is a sample mean. Its error is $\sigma_f/\sqrt{N}$, and the same draws estimate σ_f .

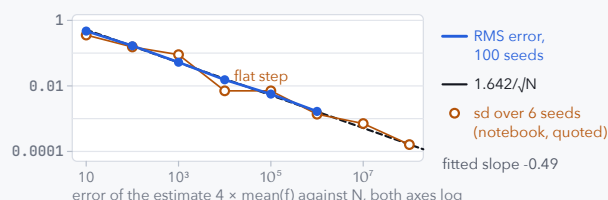
7 An expectation is an average over draws



$$\mathbb{E}_{x \sim p}[f(x)] = \int f(x)p(x) dx \approx \frac{1}{N} \sum_{i=1}^N f(x_i), \quad x_i \sim p$$

For π : $p = U([0, 1]^2)$, $f = \mathbb{1}(x_1^2 + x_2^2 \leq 1)$, $\mathbb{E}[f] = \pi/4$; the notebook's 10,000 darts give 3.1464 (quoted). The predictive is the same average, $p(y | x, \mathcal{D}) \approx \frac{1}{N} \sum_i p(y | \theta_i, x)$ with $\theta_i \sim p(\theta | \mathcal{D})$ (the Bayesian linear regression sheet). With prior draws and log-sum-exp it gives the evidence (next sheet).

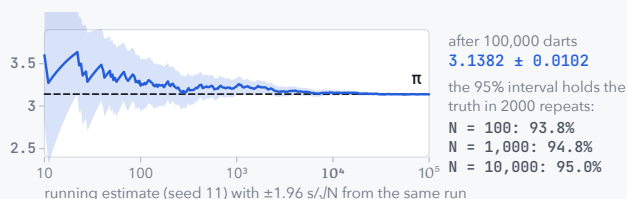
8 Unbiased, with error $\sigma_f/\sqrt{N}$



$$\mathbb{E}[\hat{\varphi}] = \varphi, \quad \text{Var}(\hat{\varphi}) = \frac{1}{N^2} \sum_{i,j} \text{Cov}(f_i, f_j) = \sigma_f^2/N$$

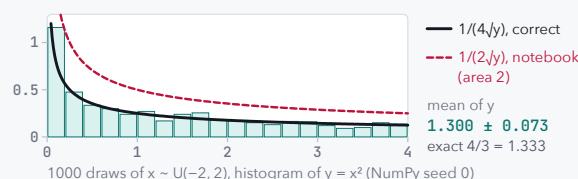
Independent draws remove the $i \neq j$ terms. For π : $\sigma_f = 4\sqrt{\pi/4(1-\pi/4)} = 1.642$. RMS error over 100 seeds: 0.472, 0.169, 0.053, 0.015, 0.0056, 0.0017 for $N = 10$ to 10^6 . The notebook's 6-seed sd stalls at 0.0071 for $N = 10^4$ and 10^5 : an sd from 6 runs falls between 0.48 and 1.49 times the truth (90% range), so the step is noise.

9 One run carries its own error bar



The sample sd s of the $f(x_i)$ estimates σ_f ; by the central limit theorem $\hat{\varphi} \pm 1.96s/\sqrt{N}$ is an approximate 95% interval; at $N = 100$ it holds π a little less often (93.8%). Halving it takes 4 times the draws. The deck derives the variance but not this interval.

10 Draws of x give draws of $g(x)$



Transform the samples and you sample $y = g(x)$, even when $p(y)$ is awkward. For $x \sim U(-2, 2)$ and $y = x^2$, each y has two roots $\pm\sqrt{y}$, so $p(y) = 2 \cdot \frac{1}{4} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{4\sqrt{y}}$ on $(0, 4]$. The course's notebook plots $1/(2\sqrt{y})$, the density for $U(-1, 1)$.

11 Uniform to normal to Monte Carlo in NumPy

```
def lcg(seed, n, a=1664525,
        c=1013904223, m=2**32):
    x, u = seed, []
    for _ in range(n):
        x = (a * x + c) % m
        u.append(x / m)
    return np.array(u)

u = lcg(42, 10_000)
x = -np.log(1 - u) / lam      # inverse CDF
cs = np.cumsum(p)            # categorical
k = np.searchsorted(cs, u, side="right")

rng = np.random.default_rng(0)
u1, u2 = rng.random((2, n))
r = np.sqrt(-2 * np.log(1 - u1))
z = np.stack([r * np.cos(2*np.pi*u2),
              r * np.sin(2*np.pi*u2)], 1)
X = mu + z @ np.linalg.cholesky(S).T

f = 4.0 * ((rng.random((N, 2))**2)
           .sum(1) <= 1)

est = f.mean()
se = f.std(ddof=1) / np.sqrt(N)
```

12 Pause and think

How many darts give π to $\pm 10^{-4}$ at 95%?
 $(1.96 \cdot 1.642/10^{-4})^2 \approx 10^9$.

Does the $1/\sqrt{N}$ rate depend on the dimension? No. A grid with k points per axis needs k^d ; but σ_f can grow with d .

Why fix the seed? Reruns repeat exactly; independent runs need different seeds.

When does the inverse CDF fail? When F^{-1} has no closed form and p is known only up to a constant, as for most posteriors: rejection and importance sampling (the next sheet).

seeded LCG $\rightarrow u \cdot x = F^{-1}(u) \cdot \text{Box-Muller } z \cdot x = \mu + Lz \cdot \mathbb{E}[f] \approx \bar{f} \pm 1.96s/\sqrt{N}$

Source: lecture slides "Sampling methods" (Monte Carlo and common distributions) and the notebooks on uniform random numbers, inverse CDF sampling, sampling the normal and Monte Carlo sampling, nipunbatra.github.io/pml-teaching. The course's LCG is rerun exactly; other draws use NumPy, fixed seeds; torch outputs are quoted. Beyond the slides: the period rule, the tests, the discrete sampler, the Box-Muller argument, the interval. Corrections: the substitution frame has $z = -\lambda y$ at $y = x$ (it is $-\lambda x$); the error frame squares outside the expectation, $\mathbb{E}[\frac{1}{N} \sum (f(x_i) - \varphi)^2]$, and its middle line does not cancel as written; the Box-Muller frame assumes $Z_0, Z_1 \sim \mathcal{N}(0, 1)$ before deriving it; the function notebook's $1/(2\sqrt{y})$ should be $1/(4\sqrt{y})$.